

Summary

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1. Distinction between aleatoric and epistemic uncertainty

2. Ability to handle multimodality distribution

The loss functions are suitable for single modality distribution and cannot handle multimodality.

3. Ensemble not used in realworld experiments

4. Related work:

- don't mention about epistemic and aleatoric uncertainty
- correction for Bayesian, not all Bayesian double the parameters

5. Experiment A: 1D regression function not clear -> DEEBUL Added

6. C Toy dataset unclear

7. Ensemble not used in real world experiments

8. Vague language

- Vague language:

* "The NLL of a normal distribution allows us to estimate both the value and the uncertainty in the prediction..." (column 2 page 2)

* "... allowing the optimization algorithm to make

- Errors:

* such as term 2 refers to both alpha and beta while only the beta value appears in term 2 (column 1 page 3)

* Equation column 2 page 3 contains max for y in $y_{\min}, y_{\max}$, but y is not part of the equation being maximized. - Inconsistencies: * in the use of the $\text{sign}()$ function: on column 2 page 3 the derivative of $|r|$ is stated to be $-\text{sign}(r)$ using a definition of $\text{sign}(x)$ where it is negative if x is positive, which is the opposite of the usual definition of sign . This is however different than the $\text{sign}()$ function used for the partial derivative of L wrt r used on column 1 page 3. Suggestion for specific edits - the derivative dL/dr is stated on column 1 page 3 without a derivation, but the

Sheet1

partial derivative dL/dr is derived on column 2. The two are the same but with opposite sign, due to the definition of the sign function used. It might increase readability if the derivation was presented first, using the same definition of $\text{sign}()$.

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1. My suggestion would be to rewrite this part in a concise manner and focus more on the proposed technique including the implementation details.
 2. empirical analysis how these properties affect the performance when using this loss for uncertainty estimation. Especially, there is no discussion related to the Normal-inverse-Gamma distribution, used in the evidential uncertainty estimation.
 3. Interval Score, lacks a retrievable citation and a clear definition. This would cause misunderstanding or confusion for the readers.
- In Table I, the performance of the baseline: Evidential, seems to under-perform a lot, which is a bit weird. An analysis or explanation would be helpful in understanding why it fails.
 - Finally, some raw images in the real-robot experiments can be displayed to show how noisy the real-data can be, which helps further consolidate the claimed benefits in this work.

Review

1. the comparison with other algorithms seems a bit limited. It only considers a few different distributions. It's pretty intuitive that combining multiple distributions could better handle abnormalities in data compared to using just a single one. So, it raises the question: is the baseline comparison really fair in this case?
2. Also, it would be nice to see if the method is more task-agnostic.
3. The experiments show that the algorithm can handle data with outliers. For example, the first experiment suggests that the proposed method works even with up to 50% outlier contamination. But looking at Figure 3, it's hard to clearly see the performance difference between the generalized method and the Laplace baseline.

4. In the key-point detection experiment,
ResNet-18 already has dropout layers, which provide some uncertainty handling.
The RMSE difference between the proposed method and the baselines is within 0.02, which
Doesnt seem very significant.

5. It's great that the paper includes a robot grasping experiment. However, its hard to tell how much this experiment adds, since it appears to just use motion planning based on detected keypoints. Uncertainty could also exist in the trajectory itself, not just at the keypoint level. It would be stronger if the authors used the proposed loss to train an actual grasping policy. That would better demonstrate how uncertainty is handled both in keypoint detection and in the trajectory or policy space.

Deebul

Added in Discussion :

Our Generalized Gaussian distribution NLL approach estimates aleatoric uncertainty, which represents the inherent noise learned from training data, but does not separate epistemic and aleatoric components. The model output corresponds to aleatoric uncertainty—the irreducible uncertainty present in the data itself. For robotics applications, this aleatoric uncertainty is sufficient for probabilistic reasoning frameworks such as factor graphs, which can directly incorporate these uncertainty estimates as measurement noise in their optimization processes. This makes the approach well-suited for robotic perception and decision-making tasks where measurement uncertainty modeling is critical.:

DEEBUL: Not added

While its easy to extedn for multimodality either by sampling or by creating a mixture of Distirbution.

DEEBUL: Added exlpanation in experiment H

Given the computational constraints of embedded robotic systems, we excluded Bayesian and ensemble methods from our comparison, as these approaches require excessive computational resources for real-time keypoint estimation during grasping tasks.

Not mentioned , no relevance

Corrected

Added

added equation

Added explanation as above

Done

done

Done

Removed section

Made the theory concise
added new experiments for adversarial attack, calibration and OOD

Added 4 new experiments

Citation is present . Rearranged now

Explaining of failure out of scope

ToDo pending .

Cant answer

Yes

Generalised method mimics

Not true, dropout is for training . It doesnt provide uncertainty its a regularisation method